▣ 01 Introduction to probability

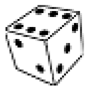
It is an everyday experience to ponder an incomplete process or a future event where there is uncertainty about the outcome. For example, there is uncertainty about:

- The numbers that will be drawn in the next lotto draw
- What the average house price will be in Ireland next year
- Whether a file server will crash in the next three days

Probability is a mechanism that allows us to quantify this uncertainty.
There are different approaches to defining probability and different ways of expressing a probability measure.

Quantifying probability

We consider the experiment of rolling a fair die and the likelihood of the <u>outcome</u> '3'.



1. We could say that the <u>chances</u> of the outcome '3' is 1 chance in 6.

2. Alternatively, we calculate the <u>probability</u> of this outcome by evaluating the proportion $^1/_6 = 0.167$

3. By scaling this last measure by 100, we obtain the probability as a <u>percentage</u> i.e., 16.7%.

4. A final measure of probability, popular in gambling contexts, involves the concept of <u>odds</u>.

   In our dice roll, there is 1 way we can get the outcome '3'.

   As there are 5 ways we can fail to get this outcome so we can describe the odds of getting a '3' as 5 to 1.

   And the odds of failing to get a '3' are 5 to 1 on.

In general, if the probability of an outcome is p then the odds are $^p/_{1-p}$.

We will usually use proportion notation, which must give us a measure in the interval [0, 1].

Note that we can readily calculate the expected number of favourable results in a given number of attempts using probability.

I.e., if we rolled a die 15 times, we might expect to get the outcome '3' on average 15(0.1667) = 2½ times.

▫ Mathematical or classical probability

Most people would be readily convinced that the first lotto ball drawn (from 45) in next week's draw will have one chance in 45 of being the number 17.

Each of the 45 possible results or <u>outcomes</u> has the same chance, as all outcomes are <u>equally favoured</u> by the selection process.

This "equally favoured" property puts this problem in the realm of mathematical probability.

Many gambling applications (rolling a die, selecting 5 cards from a pack of 52 etc.) fit into this framework.

In fact, early attempts to apply mathematics to uncertainty was prompted by the popularity of gambling.

Empirical probability and relative frequency

Select an entry at random from a phone book and consider what the first letter of the surname might be.

Certainly, there are 26 possibilities, but mathematical probability is of little use to us here.

Clearly all 26 letters are not equally favoured. An ‘M’ or an ‘O’ would be much more likely than a ‘X’ or a ‘Z’.

So how might we estimate the probability of say the letter being a ‘H’?

One way to proceed would be to take a sample of size N names from the phone book, count the number, n, of these that begin with ‘H’ and take $^n/_N$ as an empirical probability.

Of course, our confidence in our estimate would increase with N, the size of the sample we take.

In the limiting case, where we sample the entire population, we obtain a relative frequency probability.

Much uncertainty can be assigned a suitable probability measure by using the relative frequency concept.

For example, it could be used to estimate the probability that a unit in a large, mass-produced batch is defective or the probability that a member selected from the Irish population is a smoker or is unemployed or is a non-national etc.

- *Erroneously assuming (often implicitly) that any one of a finite number of possible outcomes are equally likely is a common error in the determination of a probability measure.*

Subjective probability

While the relative frequency approach is more versatile that the classical one there are still many practical situations where it is of little help.

For example, an experienced doctor may estimate that a medical intervention had a 90% chance of being successful or a knowledgeable bookmaker may assign odds of 4 to 1 on a horse winning a particular race.

It isn’t sensible to imagine events such as these as being repeated a large number of times. They are essentially one-off events and a probability estimate for an outcome represents a personal degree of belief.

Equivalence of the different approaches

We have already seen in our earlier example where we roll a die how probabilities are assigned using the classical approach.

We could arrive at the same probability measures using the empirical approach by rolling the die a number of times and observing that each of the six outcomes occur with the same frequency.

As the number of times that a die may be rolled is unlimited, this is a case where the population is infinite so we must make use of the mathematical concept of limits to apply the relative frequency notion of probability.

And of course, without ever rolling a die, we could readily postulate subjectively that all 6 outcomes are equally likely on the basis of its physical (symmetrical) properties.

Determining probability measures

Counting

In classical probability, a probability measure is obtained by taking the ratio of the number of favourable outcomes to the total number of outcomes. Thus, the calculation reduced to a problem of counting.

For example, the probability of drawing 4 cards of the same suit in a hand of 5 cards drawn from 52 is given by $^{n}/_{N}$ where n is the number of ways of drawing 5 cards from 52 in such a way that 4 of them have the same suit and the 5th has a different suit and N is the number of ways of drawing 5 cards from 52 with no restrictions.

Usually, it is not sensible to attempt to obtain such counts by enumerating every possibility (the counts can be enormous), rather we make use of various counting principles.

Calculating empirical probabilities also involves counting, but in that case the counting is in the trivial enumeration (i.e., 1, 2, 3….) sense.

Probability models

An alternative to counting is to use a modelling approach. The idea is to assume that the phenomena of interest may be represented by a suitable mathematical model taking the form of a probability density function (continuous case) or probability mass function (discrete case).

This model is then manipulated to obtain a probability measure.

Example

A simple example of such a model is the continuous uniform probability density function.

Suppose we wished to calculate the probability that a randomly selected individual will have their birthday in Jan, Feb or Mar.

We might presume that the spread of births throughout the year are evenly distributed for this population. I.e., we are imposing a continuous uniform PDF model.

We would then take our probability estimate to be $^{3}/_{12}$ or 0.25 (25%).

It might be argued that our assumed model is wrong, that the birth rate is higher in some months than in others. The reality is that all models are wrong. But they may still be useful.

The challenge is to establish if a particular model is adequate for the purpose at hand.

For example, the model above would likely be adequate as a mechanism to divide say a group of tourists on a walking tour into four roughly equal subgroups.

It may not be adequate for someone managing the resources of a maternity hospital.

We will look at some of these probability models later – including the normal, the Poisson and the binomial.

Terminology

Our die roll is an example of a random phenomenon.

Each execution of a random phenomenon is called a trial or experiment and each of these results in an outcome.

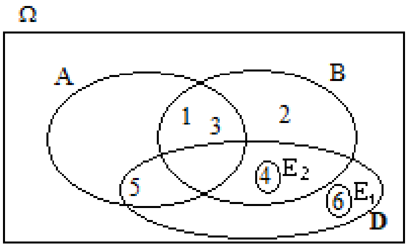
The collection of all possible outcomes is called the sample space.

A set of one or more outcomes is called an event.

A simple event consists of one outcome, otherwise the event is compound.

Simple events are conventionally represented with subscript notation E_1, E_2 etc.
Compound events with capital roman letters A, B,…

Example – a single roll of a die



Let E_1 be the outcome that a '6' results.
Let A be the event that the outcome is odd (i.e., one of 1, 3 or 5)
Let B be the event that the outcome is 4 or less (i.e., 1, 2, 3 or 4)
Let E_2 be the event that the outcome is 4.
Let D be the event that the outcome is 4 or more, i.e., 4, 5 or 6.

The event E occurs if the outcome that occurs is contained in E.
So, if the die result is '5' then events A and D above will have occurred.

$A \cap B$ is the event that both A and B occur, i.e., above, outcome is either 1 or 3

$A \cup B$ is the event that either A or B or both occurs. I.e., outcome is one of 1, 2, 3, 4 or 5

Clearly, $A \cap B = B \cap A$ and $A \cup B = B \cup A$

Ω denotes the entire sample space, i.e., an event with all possible outcomes in it. I.e., $\Omega = \{1, 2, 3, 4, 5, 6\}$

Φ denotes the null event, i.e., an event with no outcomes in it. I.e., $\Phi = \{\}$

$\overline{A}$ or $\neg A$ is the complement event of A, i.e., it is the event that A does not occur. $\overline{A} = \{2, 4, 6\}$

If $A \cap B = \Phi$ then events A and B are mutually exclusive or disjoint. Here, A and B are not.

If $A \cup B = \Omega$ then events A and B are collectively exhaustive. Again, A and B are not.

Example
Show that A and E_2 are mutually exclusive, and that B and D are collectively exhaustive.

Example
Find the event F so that A and F are mutually exclusive and collectively exhaustive.

It is easy to make errors when establishing the complement of an event. E.g., the complement of the event that the outcome from a die roll is at least 3 is the event that it is at most 2 (not 3).